## Like terms

**1** State whether the following pairs are like terms:

**a** $9y$ and $10y$ **b** $9n^2$ and $2n$ **c** $2xy$ and $4yx$ **d** $10x^2y$ and $9y^2x$

**e** $6x^3y$ and $10yx^3$ **f** $13uv$ and $17vu$ **g** $10z$ and $10b$ **h** $8y$ and $5y$

**2** Simplify the following by collecting like terms:

**a** $2a + 5a$ **b** $10x + 6x$

**c** $4b + 3b$ **d** $12y - 3y$

**e** $3c + 4c + 7c$ **f** $15x - 6x - 2x$

**g** $19b - 12b - 6b$ **h** $-3n + 6n + 3n$

**i** $8x - 3x + 7x$ **j** $20x^2 - 12x^2 + 10x^2 - x^2$

**k** $4xy + 10xy - 9xy$ **l** $10ab + 3ab - 8ba$

## Simplify expressions

**3** Consider the expression: $4y - 3y$.

**a** Simplify the expression.

**b** Explain why the answer to part (a) was not $1y$.

**4** For each of the following expressions:

**i** State the pairs of like terms. **ii** Hence, simplify the expression.

**a** $5u + 7 + 8u + 8$ **b** $7u + 2v - 7u$

**c** $3p + 5p^2 - 3p^2 + p$ **d** $4x - 6 + 5x$

**5** Simplify:

**a** $8x + 6y - 2y - 4x$ **b** $10m + 9n + 5m + 10n$

**c** $10x - 9y - 7x + 11y$ **d** $11m + 8n + 14m$

**e** $9xy + 12yx$ **f** $6p + 8q - 6p$

**g** $9p^2 - 11p + 11p^2$ **h** $7n^2 + n^2 - 4m + 2m^2 - 5n^2$

**i** $8x + 9y - 5x + 10y$ **j** $7a + 11a - 9b + b$

**k** $12x^2 + 10x - 9x - 5x^2$ **l** $8x - 7y - 6z + 10z$

**m** $15y - 8y^2 + 10y - y^2$ **n** $19a - 8b + 4c - 8a + b$

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o $2ab - 7c + ab + 9c$

p $12j + 9k - 7k + 11j$

q $18x - 9y + 7x + 11y$

r $11y - 6z + y + 6z$

s $13m + 2n - 8m + 10n$

t $-3s + 4t - 6t + 9s$

**6** Simplify:

a $5x + 2y - 8x + 6z - 3y + 4z$

b $2xy + 5yz + 4xz - 8xy + 2xz - 3yz$

c $8x^2 + 10x - 6 + 4x^2 - x + 11$

d $20 - 9x + 4y + 6y - 7x + 21$

e $9mn + 8m - 7n + 7n - 10m + 2mn$

f $-6 + 8x^2 - 9 + 5x + 16x^2$

g $11s + 6t + 3st - 2ts + t - 4s$

h $13ab - 8bc + 6ba - 9b + 11cb$

**7** Complete the following statements:

a $2x + \square = 8x$

b $7x - \square = 2x$

c $9x + 4y + \square = 9x + 6y$

d $7x + \square + 10y - \square = 11x + 6y$

e $5x^2 - 2x + \square + \square = 9x^2 + 3x$

f $8ab + \square + 9b + 6ab = \square + 11b$

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